

# COCUS Weather Project

Working as a data analyst for E-Green, one of the biggest renewables companies present in Porto, specialized in energy generation using solar and wind power, one of e-Green's objective is to improve the operational efficiency of the systems they use to produce green energy to power its office in Porto, to do so, they requested a historical analysis of Porto's weather conditions, mainly focused on temperature and wind, to understand weather patterns based on the last 20 years' data and identify trends that will allow e-Green's energy production planning team to efficiently allocate resources and define a strategy to have a sustainable, innovator office, like no other. Another requirement is to comprehend the precipitation trends over the years in Porto, this one was raised by the R&D team and this specific indicator - together with all the data insights - will allow them to adapt its new technology accordingly.

Based on what was requested by e-Green, I decided to gather data for the last 20 years, from 2002 to 2022, for the city of Porto, using Open-Meteo API. The data collected with daily granularity, included the temperature - maximum, minimums and average - in Celsius, wind speed (km/h) and the precipitation sums (mm). These resulted in more than 8000 records. The entire process until the load of the data to the database can be found on the KNIME workflow. Once the data was ready, it was loaded from the database - using a connection to Microsoft SQL server - into Spotfire, a visualization tool, to gain further insights and create informative visualizations.

The dashboard includes the following:

- Temperature variations are a fundamental variable that must be taken into consideration when designing efficient solar energy systems, furthermore, solar panels also perform better between a specific temperature range - from 25°C to 30°C, considering this, two charts were made. A line chart, to understand the evolution of the average temperature over the past 20 years, to allow e-Green to 'forecast' medium-term trends and a bar chart, to infer the average maximum temperature, per month, for the same timeline. The latter allowed us to infer how well the solar panels behave and two lines to represent peak performance intervals of these solar panels were added.

- The same rationale is applied to wind indicators, considering that there are multiple factors that influence the wind speeds for a specific location, like topography, regional climate and seasonal changes, a line chart for the average maximum wind speed throughout the years was added, as well as its average per month, over the years. These insights will allow e-Green to understand if these values are worth the investment in wind turbines or other ways of generating energy using wind.

-Analyzing the average precipitation by month for the last 20 years, was also a request from the client, and the bar chart used allows them to identify a clear pattern related with the precipitation in Porto.

The analysis shows the existence of distinct, consistent, seasonal patterns in the city of Porto. Overall, the information present on the dashboard will help e-Green to strategically plan its operations, enhancing its performance, and its developments in terms of technological advances.